Unit: Unit 0: Foundations of Algebra **Name:** ____________

Topic: Variables and Expressions / Order of Operations **Date:** __________

Homework (Jalapeno)

Q1. Evaluate the expression: $2 \cdot 3 + 12 - 8$
- (A) 10
- (B) 12
- (C) 14
- (D) 16
- (E) 18

Q2. Translate the phrase into an algebraic expression: '5 more than the product of 3 and x'
- (A) $3x + 5$
- (B) $3x - 5$
- (C) $5x + 3$
- (D) $5x - 3$
- (E) $3 + 5x$

Q3. Simplify the expression: $2x + 5x - 3x$
- (A) $4x$
- (B) $6x$
- (C) $8x$
- (D) $10x$
- (E) $12x$

Q4. Substitute x = 4 into the expression: $x^2 + 2x - 3$
- (A) 13
- (B) 15
- (C) 17
- (D) 19
- (E) 21

Q5. Evaluate the expression: $12 - 3 + 2 \cdot 4$
- (A) 13
- (B) 15
- (C) 17
- (D) 19
- (E) 21

Q6. The ancient Egyptians used the expression $5x - 2$ to calculate the area of a rectangular field. If x = 6, what is the area?
- (A) 28
- (B) 30
- (C) 32
- (D) 34
- (E) 36

Q7. The Babylonians used the expression $2x + 5$ to calculate the perimeter of a triangular field. If x = 3, what is the perimeter?
- (A) 11
- (B) 13
- (C) 15
- (D) 17
- (E) 19

Q8. The ancient Greeks used the expression $x^2 + 2x - 6$ to calculate the volume of a cube. If x = 2, what is the volume?
- (A) 0
- (B) 2
- (C) 4
- (D) 6
- (E) 8

Open Ended Questions (FRQ)

Q9. Evaluate the expression: $3 \cdot (2x + 1) - 2(x - 3)$, given x = 2

Q10. Simplify the expression: $(x + 2)(x - 3)$, and then substitute x = 4

Chef's Tips
- Remind students to use the correct order of operations
- Encourage students to read the questions carefully
- Allow students to use a calculator to check their answers